

PRACTICE MIDTERM 1

CS 410/510 — Databases

This practice exam mirrors the real Midterm 1 in structure, point split, and timing: **100 points** in **180 minutes**, five parts, the same kinds of questions in the same order. The content is entirely different. Work it under exam conditions — closed book, no computer, timed — and only then check the answer key.

Two notes on what is here that was not on previous exams. The diagram in Part A includes **many-to-many** relationships, so think carefully about what the foreign keys turn into. The data in Part C includes a row with **no match** on the other side, which matters for the last query.

Part	Points	Score
A: Parts of ER	15	
B: Data Modeling	40	
C: Queries	20	
D: Concepts	15	
E: Data Types	10	
Total	100	

Part A (15pts): Parts of an Entity-Relationship Diagram

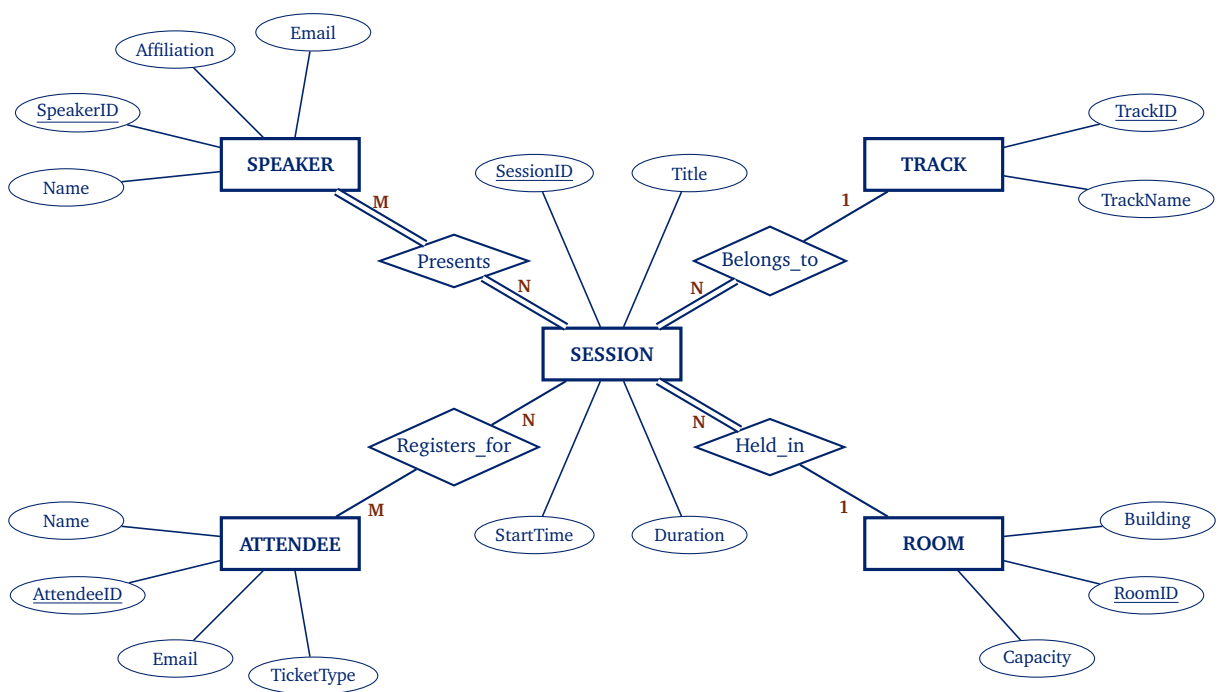
Consider the entity-relationship diagram on the next page, which models the program of an academic conference.

List the entity types:

List the relationship types:

List the primary keys:

List the foreign keys that will be needed for each table:



A **double line** marks total participation; a single line marks partial participation.

Part B (40pts): Database Modeling

In this part, you will work on a data model for a public library.

- The library catalogs **books**, which have a title, an ISBN, a publication year, and one or more authors.
- The library owns several physical **copies** of a book. Each copy has a barcode and a shelf location.
- **Members** borrow copies. A **loan** records a checkout date, a due date, and a return date — the return date is empty until the copy comes back.
- Members have a name, an e-mail address, and a membership type.
- A member can place a **hold** on a book when every copy is out.

ER Diagram (20 points)

Draw an ER diagram in Chen notation for this problem.

Relational Schema (20 points)

Describe a relational schema implementing your ER diagram for the library by listing the relations with their attributes. Identify all primary keys and foreign keys. Identify an appropriate SQL data type for each attribute.

[illegible]

Part C (20pts): Writing Queries

Consider the following tables:

Artist			Album			
artist_id	name	formed_year	album_id	artist_id	title	genre
1	Nova Trio	1998	1	2	Longitude	rock
2	The Quiet Hours	1985	2	1	Blue Arithmetic	jazz
3	Marisol Vega	2004	3	1	Second Light	jazz
4	Halvard	2012	4	3	Paper Moons	folk

Write SQL queries for each of the following (**Graduate students only** — write the queries also in relational algebra whenever possible):

Which artists formed in 1990 or later?

What are the titles of all jazz albums?

What albums does The Quiet Hours have?

What are the names of all artists who have a rock album?

How many albums does each artist have?

Part D (15pts): Concepts

Indicate whether each statement is **true** or **false**.

- _____ The schema changes often, while the instance stays fixed over the life of the database.
- _____ Every superkey is also a candidate key.
- _____ In Chen notation, a double oval marks a multivalued attribute.
- _____ A weak entity borrows part of its identity from its owner.
- _____ The relational algebra σ (select) operator picks columns from a relation.
- _____ Every relationship in an ER diagram becomes its own relation in the relational schema.
- _____ A derived attribute is stored in the database like any other attribute.
- _____ In a one-to-many relationship, the foreign key is placed on the relation for the 'one' side.
- _____ A primary key may contain NULL values as long as it is unique.
- _____ The natural join ($\bowtie$) can be expressed using product, select, and project.
- _____ A multivalued attribute can be stored directly as a single column in a relation.
- _____ Total participation is drawn as a double line in Chen notation.
- _____ SQL is declarative: it describes what to retrieve, not how to retrieve it.
- _____ Two different ER diagrams can map to the same set of relations.
- _____ In the relational model, the order of the rows in a relation is significant.

Part E (10pts): Data Types

Select the most appropriate data type (from the right column) for each item.

1. _____ The number of seats in a lecture hall
2. _____ A course catalog description
3. _____ A student's photo ID image
4. _____ The exact price of a textbook
5. _____ A building's street address
6. _____ The date a library book is due
7. _____ The precise moment a payment was processed
8. _____ The time a daily backup job starts
9. _____ A sensor's measured humidity reading
10. _____ A department's abbreviation (e.g. "CS")

Choices

- A. INTEGER
- B. DECIMAL
stores exact decimal numbers
- C. REAL
stores floating point approximations of real numbers
- D. VARCHAR
stores short, variable-length strings
- E. TEXT
stores long text, up to 65,535 characters
- F. BLOB
stores Binary Large Objects up to 65,535 bytes of data
- G. DATE
- H. TIME
- I. TIMESTAMP
stores date and time together